

0.1 Section 2.4 - Sequences and Series

Definition 1 (Sequence, Explicit Formula / nth term)

A **sequence** is an ordered list of numbers

$$a_1, a_2, a_3, \dots$$

Equivalently, a sequence is a function $f : \mathbb{Z}^+ \rightarrow \mathbb{R}$

We call f the **explicit formula** or **nth term** of a sequence

Example 1

Determine the next 3 terms of the following sequences and the explicit formula

$$1. \quad 5 \xrightarrow{+7}, 12 \xrightarrow{+7}, 19 \xrightarrow{+7}, 26 \xrightarrow{+7}, \dots, 33 \xrightarrow{+7}, 44 \xrightarrow{+7}$$

Solution:

$$\begin{aligned} a_n &= 7n + b \\ 5 = a_1 &= 7(1) + b \\ b &= 7 \\ \boxed{a_n &= 7n - 2} \end{aligned}$$

$$2. \quad 7 \xrightarrow{\cdot 3}, 21 \xrightarrow{\cdot 3}, 63 \xrightarrow{\cdot 3}, 189 \xrightarrow{\cdot 3}, \dots, 567 \xrightarrow{\cdot 3}, 1701 \xrightarrow{\cdot 3}, 5103 \xrightarrow{\cdot 3}$$

Solution:

$$\begin{aligned} a_n &= c3^n \\ a_1 &= 7 = c3^1 \\ c &= \frac{7}{3} \\ a_n &= \frac{7}{3}3^n \\ \boxed{a_n &= 7 \cdot 3^{n-1}} \end{aligned}$$

3. 1, 2, 6, 24, 120, 720, 5040

$a_n = n!$, read as "n factorial"

$$0! = 1$$

4. 1, 1, $\underbrace{2}_{1+1}$, $\underbrace{3}_{2+1}$, $\underbrace{5}_{3+2}$, $\underbrace{8}_{5+3}$, $\underbrace{13}_{8+5}$, $\underbrace{21}_{13+8}$,

Solution:

$$f_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right] \quad (\text{We'll derive this later})$$

Another solution is

$$a_1 = 1$$

$$a_2 = 2$$

$$a_n = a_{n-1} + a_{n-2}, n \geq 3$$

Definition 2

A **recursive** sequence is defined in terms of the previous terms of the sequence

Example 2

Here are some examples of recursive sequences

1. $a_1 = 2, a_2 = 3, a_n + 1 = 5a_n - 6a_{n-1}, n \geq 2$

$$n = 2, a_3 = 5a_2 - 6a_1 = 5 \cdot 3 - 6 \cdot 2 = 15 - 12 = 3$$

$$n = 3, a_4 = 5a_3 - 6a_2 = 5 \cdot 3 - 6 \cdot 3 = 15 - 18 = -3$$

$$n = 4, a_5 = 5a_4 - 6a_3 = 5 \cdot (-3) - 6 \cdot 3 = -15 - 18 = -33$$

2. $a_1 = 1, a_2 = 2, a_3 = 4, a_n = a_{n-1} + a_{n-2} - n - 3, n \geq 4$

$$a_4 = 4 + 2 - 1 = 5$$

$$a_5 = 5 + 4 - 2 = 7$$

$$a_6 = 7 + 5 - 4 = 8$$

Definition 3 (Series)

A **series** is the sum of an ordered list of numbers

$$a_1 + a_2 + a_3 + \dots$$

$$\sum_{k=1}^{\infty} a_k$$

Where:

k = Index

1 = Starting value

∞ = Ending value

$a_k = k^{\text{th}}$ term

Example 3

Find the sum of the series

$$1 + 2 + 3 + \dots + 100$$

Solution:

$$1 + 2 + 3 + \dots + 98 + 99 + 100$$

$$(1 + 100) + (2 + 99) + (3 + 98) + \dots$$

$$(101) + (101) + (101) + \dots$$

Since there's 100 things we need to add, there's 50 pairs

So

$$\text{sum} = 101(50)$$

$$\boxed{= 5050}$$

Example 4

But what about in general?

$$1 + 2 + 3 + \dots + n = ?$$

We have two cases:

1. When n is even
2. When n is odd

If we assume n is even, then

$$\boxed{1 + 2 + 3 + \dots + n = (n + 1) \cdot \frac{n}{2}} \quad (1)$$

Since $(n + 1)$ gives the value of each pair and $\frac{n}{2}$ gives the number of pairs

If we assume n is odd, then

$$\begin{aligned} [1 + 2 + 3 + \dots + (n - 1)] + n &= n \cdot \frac{n - 1}{2} \\ &= \frac{n^2 + n}{2} \\ &\boxed{= (n + 1) \cdot \frac{n}{2}} \end{aligned}$$

since n is the value of each pair $(n - 1 + 1)$ and $\frac{n-1}{2}$ is the number of pairs

Another way to think about it is to let S represent the sum

$$\begin{array}{r}
 S = 1 + 2 + 3 + \dots + (n-2) + (n-1) + n \\
 S = n + (n-1) + (n-2) + \dots + 3 + 2 + 1 \quad \text{(Switch Order)} \\
 \hline
 2S = \underbrace{(n+1) + (n+1) + (n+1) + \dots + (n+1) + (n+1) + (n+1)}_n \quad \text{(Add)} \\
 \boxed{S = (n+1) \cdot \frac{n}{2}}
 \end{array}$$

Writing this in summation notation

$$\sum_{k=1}^n \frac{n(n+1)}{2}$$

To Review...

1.

$$\sum_{k=1}^n c = \underbrace{c + c + \dots + c}_n = \boxed{nc} \quad \text{(North Carolina Rule)}$$

2.

$$\sum_{k=1}^n k = 1 + 2 + 3 + \dots + n = \boxed{(n+1)\frac{n}{2}} \quad \text{(Pairing Rule)}$$

3.

$$\sum_{k=1}^n k^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \boxed{\frac{n(n+1)(2n+1)}{6}} \quad \text{(Prove Later)}$$

4.

$$\sum_{k=1}^n k^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \boxed{\left[\frac{n(n+1)}{2} \right]^2} \quad \text{(Prove Later)}$$

You **MUST** know these 4! They will be on exams!

Example 5

Compute the Following

1.

$$\sum_{k=1}^6 k = \frac{6}{2}(6+1) = \boxed{21} \quad (\text{Use formula})$$

2.

$$\begin{aligned} \sum_{k=1}^6 2k &= (2 \cdot 1) + (2 \cdot 2) + (2 \cdot 3) + (2 \cdot 4) + (2 \cdot 5) + (2 \cdot 6) \\ &= 2(1 + 2 + 3 + 4 + 5 + 6) \\ &= 2 \sum_{k=1}^6 k \\ &= 2 \cdot 21 \\ &= \boxed{42} \end{aligned}$$

Note

This means that we can **factor** in sums!

3.

$$\sum_{k=1}^6 2 = 6 \cdot 2 = \boxed{12} \quad (\text{Use formula})$$

4.

$$\begin{aligned} \sum_{k=1}^6 (k+2) &= (1+2) + (2+2) + (3+2) + (4+2) + (5+2) + (6+2) \\ &= (1+2+3+4+5+6) + (2+2+2+2+2+2) \\ &= \sum_{k=1}^6 k + \sum_{k=1}^6 2 \\ &= 21 + 12 \\ &= \boxed{33} \end{aligned}$$

Note

This means that in general,

$$\sum_{k=1}^n c a_k = c \sum_{k=1}^n a_k \quad (\text{Factor out constants})$$

$$\sum_{k=1}^n (a_k + b_k) = \sum_{k=1}^n a_k + \sum_{k=1}^n b_k \quad (\text{Split})$$

5.

$$\sum_{k=1}^{200} (k+1)^2 = \sum_{k=1}^{200} (k^2 + 2k + 1) \quad (\text{Expand})$$

$$= \sum_{k=1}^{200} k^2 + 2 \sum_{k=1}^{200} k + \sum_{k=1}^{200} 1 \quad (\text{Split up})$$

$$= \frac{200(201)(401)}{6} + 2 \cdot \frac{200(201)}{2} + 200 \cdot 1 \quad (\text{Use formulas})$$

$$\boxed{= 2727100} \quad (\text{Use Calculator})$$

We could also do

$$\begin{aligned} \sum_{k=1}^{200} (k+1)^2 &= 2^2 + 3^2 + 4^2 + \dots + 201^2 \\ &= (1^2 + 2^2 + 3^2 + 4^2 + \dots + 201^2) - 1^2 \\ &= \frac{201(202)(403)}{2} - 1 \end{aligned}$$

$$\boxed{= 2727100} \quad (\text{Use Calculator})$$

Theorem 1 (Constant Shift)

This means that when you have a **constant shift** like $(k+1)$ is a shift from k , then

$$\sum_{k=1}^n f(k+a) = \sum_{k=1}^{n+a} f(k) - \sum_{k=1}^a f(k)$$

or

$$\sum_{k=m}^n a_k = \sum_{k=1}^n a_k - \sum_{k=1}^{m-1} a_k$$

Where $f(k)$ is a function like k^2 that has it's own formula for the sum like

$$\sum_{k=1}^n f(k) = \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Using this on the other example,

$$\begin{aligned} \sum_{k=1}^{200} (k+1)^2 &= \sum_{k=1}^{201} k^2 - \sum_{k=1}^1 k^2 \\ &= \frac{201(202)(403)}{2} - \frac{1(2)(1)}{2} \\ &= 2727101 - 1 \\ &= 2727100 \end{aligned} \quad \text{(Use Calculator)}$$

6.

$$\sum_{k=1}^{10} 2^k = S$$

$$S = 2^1 + 2^2 + 2^3 + \dots + 2^9 + 2^{10}$$

$$2S = 2^2 + 2^3 + 2^4 + \dots + 2^{10} + 2^{11}$$

(Subtract)

$$S - 2S = 2^1 - 2^{11}$$

$$S(1 - 2) = 2^1 - 2^{11}$$

$$S = \frac{2 - 2^{11}}{1 - 2} = \frac{2 - 2048}{-1}$$

$$= 2046$$

Note

This means that

$$S = \sum_{k=0}^n r^k \quad (\text{When } k \text{ starts at } 0)$$

$$S = 1 + r^1 + r^2 + \dots + r^{n-1} + r^n$$

$$rS = 0 + r^1 + r^2 + r^3 + \dots + r^{n-1} + r^n + r^{n+1}$$

(Subtract)

$$S - rS = 1 - r^{n+1}$$

$$S(1 - r) = 1 - r^{n+1}$$

$$\boxed{S = \frac{1 - r^{n+1}}{1 - r}}$$

$$S = \sum_{k=1}^n r^k \quad (\text{When } k \text{ starts at } 1)$$

$$S = r^1 + r^2 + \dots + r^{n-1} + r^n$$

$$= r(1 + r^1 + r^2 + \dots + r^{n-2} + r^{n-1})$$

$$= r \sum_{k=1}^{n-1} r^k$$

$$= r \cdot \frac{1 - r^{n-1+1}}{1 - r}$$

$$\boxed{= r \cdot \frac{1 - r^n}{1 - r}}$$

In general,

$$\boxed{\sum_{k=\text{start}}^n r^k = \frac{r^{\text{start}} - r^{n+1}}{1 - r}}$$

Note

This works for $r \in \mathbb{R}, (-\infty, \infty)$, including **negative** and **fractions**!

This can also work for sums going to ∞ , remember that a fraction to ∞ is just 0

There's more examples, but I'm too lazy to write them, so I'll just write the equations

Note

$$\sum_{k=0}^{\infty} kr^k = \sum_{k=1}^{\infty} kr^k = \boxed{\frac{r}{(1-r)^2}}$$